João Pedro Dudziak Fonseca

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EDUCATION

Georgia Institute of Technology, Atlanta, GA — Graduation: *Dec 2026*

- BS Mechanical Engineering (Automation/Robotic Systems Conc.), CS Minor (Computing/Intelligence), GPA: 3.89/4.00

EXPERIENCE

Vehicle Dynamics & Modeling Engineering Intern at Tesla, Palo Alto, CA — Jan 2026 - Jun 2026

- Analyzed competitive benchmark tests; created hysteresis, steering stiffness and yaw recovery metrics to analyze Dynamic Torque Vectoring (DTV) and Electronic Limited-Slip Differential (eLSD) activation at saturation for different drive modes.
- Developed the standard maneuver test procedure and analysis end-to-end for Stepsteer, Impact, Sinusoid, and Rampsteer, including ISO-compliance, maneuver extraction, data filtering/processing, metric calculations, plots and theoretical insights.
- Re-built all I/O infrastructure of Data Analysis Tool including parsing MoTeC/Dewesoft/Modelica data, ISO filtering/delay, unit conversion, sign/location transformations, and DataFrame validation with Pydantic classes across 130+ files (50k LoC).
- CAD-designed in CATIA / K&C tested thermal-isolator snubbers, suspension spring pads and strut tower braces to 10x lateral stiffness for a vehicle program, including hand-calculations, results-validation, 3D printing, FEA, CAD drawings and DFMA.
- Built a Python shape-constrained spline-fitting library (5k LOC) with parametric fitting used in all maneuver analysis pipelines.
- Built ImGui app for all multibody-dynamics (MBD) simulation workflows, including Vehicle/Controller YAML editing, FMU generation, MicroSim execution, database upload, and DAT metric and characteristic plot integration.

Underwater Jet Propulsion Researcher at Bhamla Lab (DARPA Funded), Atlanta, GA — May 2025 - Dec 2025

- Conducted experimental/computational fluid dynamics research on self-designed, squid-inspired flexible nozzle to improve underwater efficiency beyond conventional propellers (η = 60% to > 80%).
- 3D Particle Tracing Velocimetry (PTV) with high-speed and laser-based visualization to track flow and nozzle deformation.
- Validated Fluid-Structure Interaction (FSI) simulation models by comparing experimental flow to numerical predictions, contributing to the development of a Physics-Informed Neural Network (PINN) optimization framework for nozzle shape.

Equipment Engineering Intern at Coca-Cola (2 Rotations), Atlanta, GA — May 2023 - Aug 2023, Jan 2024 - May 2024

- Researched and integrated palm recognition payment in vending machines, estimated to cut customer journey time by 10%. Secured a partnership for the technology, kickstarting a company worldwide project, now in market trials.
- Designed and 3D printed custom bottles with standard threading to test carbonation in dispensed liquids, automating current methods by using a vibrating table. Cut test-time in half; improved measurement accuracy by 2 decimal places.
- Developed the Technical Authorization Requirements (TAR) for 20k+ deployed Coke&Go coolers.
- Computed tipping angles with added forces through manual calculations and FEA for the new line of solar-powered vending machines, validating stability with factor of safety of 2 and adherence to ISO safety standards.

Aerodynamics & Chassis Engineer at GT FSAE HyTech Racing, Atlanta, GA — Sep 2022 - May 2026

- Designed suspension wishbones for car in CATIA; validated aerodynamic behavior with Fluent and wind tunnel testing.
- Adjusted the front-wing endplate’s angle of attack by 3 degrees based on F1 aerodynamic literature, increasing downforce by 4 kg at 120 km/h with negligible drag penalty, improving stability in high-speed cornering.
- Developed / validated chassis with 3D FEA in Ansys / hand-calculations to prevent frame failure encountered by 2022’s car.

Production Engineering Intern at Sorvetes Rochinha Ice Cream CO, São Paulo, Brazil — *May 2022 - Aug 2022*

- Applied process mapping to the entire production line and identified packaging as a bottleneck.
- Reduced packaging prep time by 1 hour (per 8 hour shift) by designing a mount for an automatic labelling machine.
- Investigated scrap on industrial popsicle machine. Identified a wrapping problem preventing heat sealing in +10% of units.

PROJECTS

Founder & Lead Engineer at TrailMate (Follower Robot Backpack), Atlanta, GA — Apr 2025 - Present

- Built an autonomous robot, including powertrain (geared DC motor with 1:9 transmission), steering (servo-driven rack and pinion), suspension (double-wishbone), using CAD / FEA stress and buckling analysis in NX to support 20 kg+ payloads.
- Developed a real-time control pipeline on Raspberry Pi and STM32 with OpenCV obstacle avoidance and NanoDet neural network person tracking, implementing a scoring algorithm to compute steering / throttle commands based on relative position.

Inverse Kinematics Approximation for Robotic Arms with Machine Learning, Atlanta, GA — May 2025 - Aug 2025

- Developed a machine learning framework for robotic arm inverse kinematics, achieving sub-degree joint accuracy and sub-millisecond inference for real-time, high-DOF manipulators; awarded “Best Project” among 30+ teams.
- Built supervised (MLP, Gradient Boosting) / unsupervised (K-Means, DBSCAN) models with PyTorch / Scikit-Learn for reachability and joint-space regression on 1.5M+ PyBullet samples; 10x precision and speed over analytical IK solvers.

SKILLS & HONORS

Prototyping Instructor (PI) at Georgia Tech Invention Studio - 2024
UKMT Mathematics Challenge Gold Medalist - 2020 | 2021
Engineering: SolidWorks, CATIA, Siemens NX, Blender, Ansys, COMSOL, Dymola, Simulink, PyBullet, CFD, FEA, Thermal, Stress Analysis, Mechanical Design, Product Development, Design Validation, GD&T, DFMA, BOMs, CNC Machining, 3D Printing, Microcontrollers, Vehicle Dynamics, Mechanics of Materials, NVH
Coding: Python, MATLAB, Java, C/C++, Modelica, HTML, PyTorch, ScikitLearn, React, STM32, Raspberry Pi, Pandas, NumPy, Pydantic, Git, SciPy, data pipelines